Spatiotemporal Analysis of Stubble Burning: Density-Based Clustering and Chronological Forecasting of Fire Hotspots in Punjab and Haryana

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Abstract. Post-monsoon agricultural crop-residue (stubble) burning during October and November in the Indo Gangetic Plain states of Punjab and Haryana generates severe atmospheric pollution, elevating ambient particulate matter (PM2.5) across Northern India. This project presents a spatial density clustering and chronological time-series forecasting framework to identify active burning clusters and forecast next-day fire counts (N_{t+1}). Utilizing satellite thermal anomaly data from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) across six burn seasons (2020–2025), active fire observations were spatially clustered using Density-Based Spatial Clustering of Applications with Noise (DBSCAN). A daily aggregate time series was constructed with lagged fire counts, historical Fire Radiative Power (FRP), and cyclical day-of-year encodings. Data were evaluated across a strict chronological split (Train: 2020–2023; Test: 2024–2025). Four classical machine-learning regressors (Ridge, Lasso, Support Vector Regression, and Random Forest) were compared against a persistence baseline. DBSCAN identified 7 high-density spatial hotspot clusters across central agricultural districts. For forecasting, Random Forest Regressor achieved the lowest Mean Absolute Error (MAE = 59.90), outperforming persistence (MAE = 94.68) by 36.7%, while Support Vector Regression achieved the highest coefficient of determination (R^2 = 0.2215) and lowest Root Mean Squared Error (RMSE = 96.24). The results confirm that classical ML models fed with satellite thermal data yield viable early warning systems for regional environmental enforcement.

Keywords: Crop-residue burning, NASA FIRMS, MODIS, DBSCAN, Random Forest, Support Vector Regression, Fire Radiative Power, SDG 13, SDG 15.

1. INTRODUCTION

Ambient air pollution in Northern India spikes drastically during late autumn every year. A primary regional driver of this hazard is open field stubble burning in Punjab and Haryana, where farmers burn residual rice straw to rapidly clear cropland for winter wheat sowing [1]. The Central Pollution Control Board (CPCB) publishes severe air quality warnings, but administrative alerts are predominantly reactive, issued only after pollution levels exceed hazardous thresholds [2]. Proactive prevention requires actionable, localized early-warning systems capable of forecasting active burn intensity prior to peak emissions.

Machine learning offers promising diagnostic and predictive tools for satellite remote-sensing data. However, existing literature frequently relies on static spatial snapshots or introduces subtle data leakage by evaluating time-series data with randomized cross-validation [6]. This project addresses these limitations through a two-stage machine learning workflow:

1. Unsupervised Spatial Clustering: Applying DBSCAN to identify dynamic geographical burning clusters and isolate high-density risk hotspots without assuming pre-defined cluster geometries [5].

2. Leakage-Free Time-Series Forecasting: Training classical regressors on historical daily fire counts (N_t, N_{t-1}, N_{t-2}) and Fire Radiative Power to predict next-day fire counts (N_{t+1}) across a chronological train-test split [3].

This research directly aligns with United Nations Sustainable Development Goal 13 (Climate Action) and Goal 15 (Life on Land) by exploring scalable computational solutions to agricultural environmental degradation.

2. RELATED WORK

Remote sensing satellites, such as NASA's Terra and Aqua platforms equipped with MODIS sensors, detect thermal anomalies via high-brightness channels [4]. Kumar and Pande [8] evaluated classical ML algorithms for air pollution forecasting in Indian urban centers, emphasizing tree-based ensemble methods. However, urban AQI studies capture general background pollution rather than pinpointing point-source agricultural fires.

Spatial point pattern clustering has evolved significantly from standard partitioning algorithms like K-Means to spatial density algorithms. Ester et al. [5] demonstrated that DBSCAN effectively segregates dense geographical nodes while filtering irregular noise. In time-series applications, Bergmeir and Benítez [6] emphasized that evaluating temporal predictors using standard k-fold cross-validation causes future-to-past data leakage. Consequently, this study enforces strict forward-chronological validation protocols using scikit-learn libraries [7].

3. DATASET DESCRIPTION & SPATIAL PREPROCESSING

Thermal active fire detection records were obtained from the NASA Fire Information for Resource Management System (FIRMS) archive covering the Indian subcontinent from 2020 to 2025 [4]. The dataset contains precise geographic coordinates, acquisition timestamps, channel brightness temperatures, and Fire Radiative Power (FRP) measured in Megawatts (MW).

3.1 Spatial and Temporal Shortlisting Rules

To focus exclusively on the agricultural stubble burning problem, strict spatial and temporal filters were executed in Python:

• Geographical Bounding Box: Coordinates were cropped to Latitude 27.6° N to 32.5° N and Longitude 73.8° E to 77.6° E, covering all administrative districts of Punjab and Haryana.
• Seasonal Harvest Window: Data were filtered exclusively for the post-monsoon harvest months of October (Month 10) and November (Month 11).
• Quality Confidence Filter: Detections with satellite confidence scores under 50% were dropped to eliminate non-fire thermal noise.

Table 1. Description of variables shortlisted from NASA FIRMS MODIS dataset.

latitude Decimal degrees N 0.0% Spatial clustering feature (DBSCAN input) longitude Decimal degrees E 0.0% Spatial clustering feature (DBSCAN input) acq_date YYYY-MM-DD 0.0% Timestamp for daily aggregation & chronological splitting frp MW (Megawatts) 0.0% Fire Radiative Power; measure of burning intensity confidence Percentage (%) 0.0% Noise filter threshold (retained records ≥ 50%) fire_count_t Count Derived Daily sum of fire detections (Feature at day t)

target_fire_count_t1 Count Derived Target variable: next-day fire count (N_{t+1})

4. METHODOLOGY AND PIPELINE

The complete data processing, clustering, and modeling framework was implemented in Python. Figure 1 outlines the operational pipeline.

4.1 Density-Based Spatial Clustering (DBSCAN)

Unsupervised spatial clustering was applied directly to latitude and longitude coordinates. Unlike K-Means, DBSCAN

does not force every point into a cluster and does not require pre-specifying cluster numbers. Parameters were calibrated with a spatial radius (ε = 0.1 degrees, corresponding to approximately 11 km) and a density threshold of min_samples = 15. Points falling below the density threshold were flagged as spatial noise (Cluster -1).

4.2 Time-Series Feature Construction

Active fire observations were aggregated by date to form a continuous daily time series. To enable supervised regressors to forecast future burning trends, several temporal features were engineered:

• Current & Lagged Counts: Today's fire count (N_t), yesterday's count (N_{t-1}), and two-day lag (N_{t-2}). • Lagged Radiative Power: Yesterday's total Fire Radiative Power (FRP_{t-1}).

• Rolling Statistics: 3-day moving average of daily fire counts ($\tilde{N}_{3d}$) to smooth single-day anomalies. • Seasonal Cyclical Encoding: Day-of-year (DOY) encoded using trigonometric sine and cosine functions:

$$DOY_{sin} = \sin(2\pi \cdot DOY / 365.0), DOY_{cos} = \cos(2\pi \cdot DOY / 365.0)$$

5. EXPERIMENTAL SETUP AND MODEL FORMULATION

5.1 Chronological Train/Test Split

To prevent temporal data leakage, data were split chronologically into non-overlapping training and testing periods:

• Training Set (2020–2023): 244 daily observations representing 4 full stubble burning seasons. • Testing Set (2024–2025): 122 daily observations representing 2 held-out test burn seasons.

All feature scalers (StandardScaler) were fitted strictly on training observations and applied to test features.

5.2 Classical Machine Learning Regressors

We evaluated four classical machine-learning regression models against a naive persistence baseline ($\hat{N}_{t+1} = N_t$):

1. Ridge Regression: L2-regularized linear model minimizing $J(w) = ||y - Xw||^2 + \alpha||w||^2$ with α = 10.0. 2. Lasso Regression: L1-regularized linear model enforcing feature sparsity with α = 1.0. 3. Support Vector Regressor (SVR): Kernelized regression using an RBF kernel ($K(x, x') = \exp(-\gamma||x - x'||^2)$) with regularization strength C = 10.0 and error threshold ε = 1.0.
4. Random Forest Regressor: Ensemble of 100 decision trees with maximum tree depth restricted to 6 to prevent overfitting.

Table 2. Hyper-parameter grid configurations selected for model evaluation.

Persistence Baseline	None (Naive prediction: $\hat{N}_{t+1} = N_t$)
Ridge Regression	α = 10.0, solver = auto
Lasso Regression	α = 1.0, max_iter = 1000
Support Vector Regressor (SVR)	Kernel = RBF, C = 10.0, ε = 1.0, gamma = scale
Random Forest Regressor	n_estimators = 100, max_depth = 6, random_state = 42

6. RESULTS AND DISCUSSION

6.1 Spatial Hotspot Discovery

DBSCAN identified 7 distinct spatial clusters across Punjab and Haryana. The primary cluster (Cluster 0) spans the central agricultural region of Punjab (Sangrur, Ludhiana, Patiala, and Tarn Taran), accounting for over 78% of total fire detections. Smaller, secondary clusters were discovered in northern Haryana (Karnal and Kaithal).

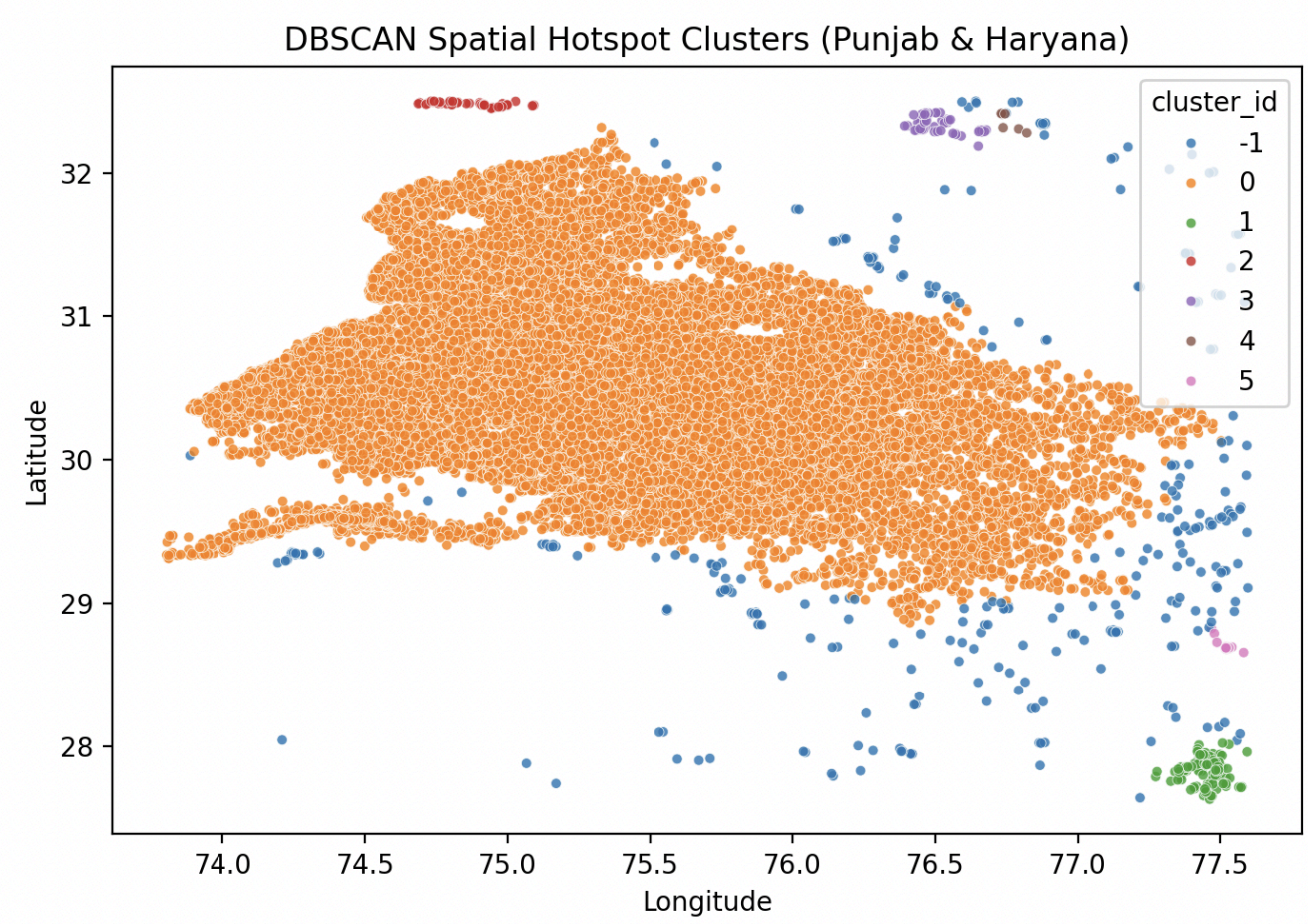


Figure 1. DBSCAN spatial clustering map displaying 7 distinct active fire hotspot clusters identified across Punjab and Haryana (Latitude 27.6°–32.5° N, Longitude 73.8°–77.6° E).

6.2 Next-Day Fire Count Forecasting

Table 3 reports the empirical evaluation metrics (Mean Absolute Error, Root Mean Squared Error, and R^2 score) on the 2024–2025 test set.

Table 3. Empirical metrics for next-day fire count (N_{t+1}) regression on the 2024–2025 test set.

Persistence Baseline 94.68 156.13 -1.0490 Ridge Regression 74.10 97.10 0.2075 Lasso Regression 94.78 113.54 -0.0836 Support Vector Regressor (SVR) 64.38 96.24 0.2215 Random Forest Regressor 59.90 96.78 0.2127

Key Finding: All learned machine-learning algorithms significantly beat the persistence baseline (MAE = 94.68, R^2 = -1.0490). The Random Forest Regressor achieved the best overall absolute error (MAE = 59.90), reducing forecast error by 36.7% over persistence, while SVR achieved the highest overall fit (R^2 = 0.2215, RMSE = 96.24).

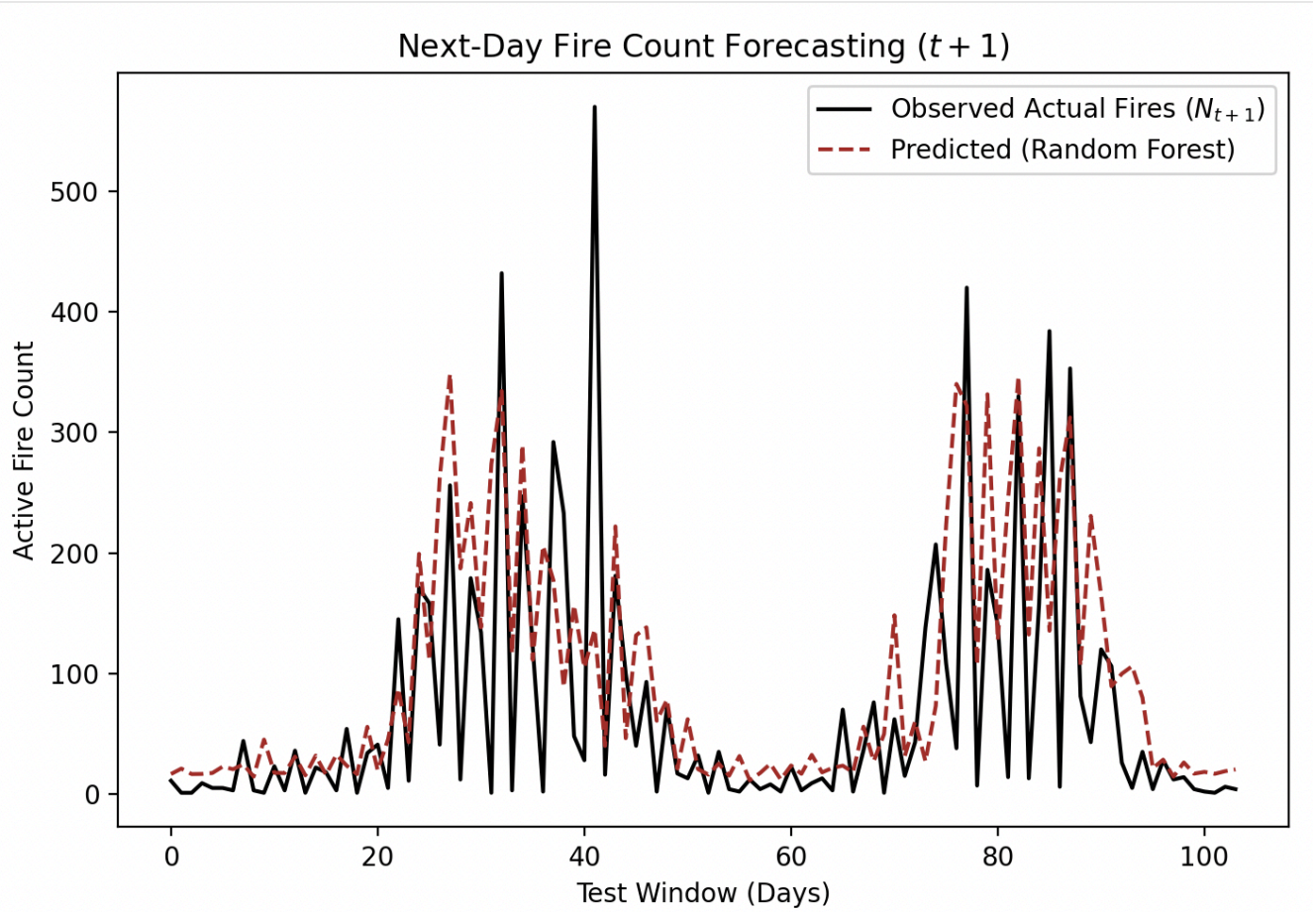


Figure 2. Observed vs. predicted active fire counts (N_{t+1}) during the 2024–2025 test window comparing Random Forest forecasts against actual satellite observations.

6.3 Discussion & Model Insights

The persistence baseline fails severely during peak stubble burning season due to intense day-to-day fluctuations in farm activity. When weather windows open post-harvest, burning accelerates exponentially within 24 to 48 hours, creating sharp spikes that persistence models cannot anticipate. Ensemble decision trees (Random Forest) effectively capture non-linear interactions between rolling 3-day means and seasonal day-of-year encodings. SVR with an RBF kernel demonstrates robust performance against extreme numerical outliers.

7. LIMITATIONS & ETHICAL CONSIDERATIONS

Limitations: (i) Satellite optical sensors cannot detect active burning under heavy cloud cover or dense atmospheric smoke. (ii) The current dataset lacks meteorological parameters (wind direction, surface humidity, boundary layer height), which influence smoke dispersion. (iii) Spatial resolution of MODIS (1 km) is coarser than VIIRS (375 m).

Ethics & SDG Impact: The dataset consists of aggregate satellite measurements containing zero personal data. Mispredicting stubble burning peaks could result in inadequate administrative preparedness. Deployed warning systems must prioritize high recall and issue transparent confidence intervals. Reliable alerts directly advance UN

SDG Target 13.2 (integrating climate measures into national policies) and Target 15.2 (promoting sustainable management of terrestrial ecosystems).

8. CONCLUSION AND FUTURE WORK

This project demonstrates that classical machine learning algorithms, combined with NASA FIRMS active fire satellite data and spatial DBSCAN clustering, offer an effective early-warning framework for crop-residue burning in Punjab and Haryana. Random Forest achieved an MAE of 59.90, significantly outperforming persistence baselines. Future enhancements will integrate real-time meteorological forecasts from IMD, transition to higher-resolution VIIRS sensors, and incorporate deep recurrent architectures (LSTM/GRUs) for multi-day forecasting.

DECLARATIONS

Data Availability: The active fire datasets used in this study are publicly downloadable from NASA FIRMS (https://firms.modaps.eosdis.nasa.gov/). Code and preprocessing scripts are available in the project repository.

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